Airavata Group Poster

Students: Jiahao Chen, Yu Miao, Hongyi Wang
Mentors: Tyler Hayes, Giri Krishnan, Suresh Marru

Georgia Tech
Open Source Program Office

Project Overview

1. AI4SCIENCE – VizFold: The **VizFold project aims to develop** tools to extract, visualize, and manipulate AlphaFold's internal protein representations. **From** creating scalable pipelines and web interfaces to visualize AlphaFold embedding to reproduce and generate novel Evoformer embeddings. Together, these projects provide both interpretability tools for understanding AlphaFold's internal workings and generative capabilities for creating new protein structures.
2. MCP for Cybershuttle: This open source project integrates a LangChain-based AI Agent with the Apache Airavata and Cybershuttle platforms via the Model Context Protocol (MCP). The goal is to create an intelligent, conversational interface that can query, retrieve, and interact with scientific resources such as datasets, repositories, and projects from the Cybershuttle research catalog.

Goals and Milestones

AI4SCIENCE: on one hand, it takes a generative approach, aiming to build autoencoder models to reproduce and resample Evoformer embeddings, enabling the generation of novel protein structure representations by learning to navigate AlphaFold's latent space. Project 1 focuses on visualizing AlphaFold embeddings. It includes cloning and testing OpenFold locally, scripting batch inference and clustering (PCA/UMAP + K-means/DBSCAN), evaluating with clustering metrics, building an offline Streamlit viewer, scaling on HPC, and ultimately enabling real-time-sequence inference and visualization via Streamlit.

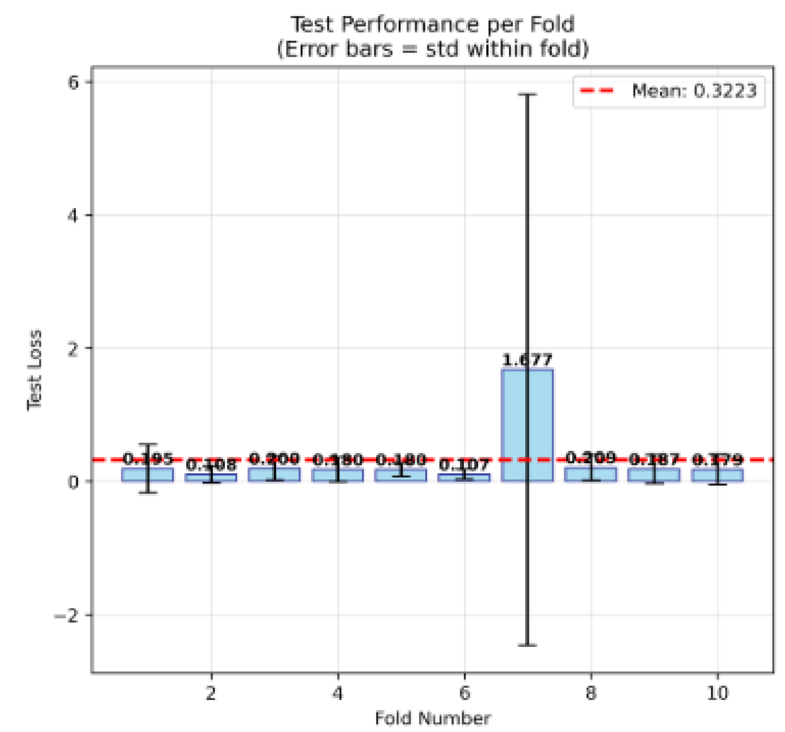
MCP: MVP Agent running on Cybershuttle Jupyter with functioning endpoints. LangChain tools: list_resources, get_all_tags, create_repository, etc. MCP client hosted within airavata-portals-mcp-client-fork

Open Source Outcomes

All code and models are released as open-source resources for the community. AI4Sci's repo produced open-source utilities for extracting and processing OpenFold Evoformer embeddings, including standardized HDF5 storage formats and preprocessing scripts that can benefit the broader computational biology community in parallel with OpenFold. While MCP's client chatbox front and back ends are in airvata and cyber shuttle's repository, with demos and sample usages.

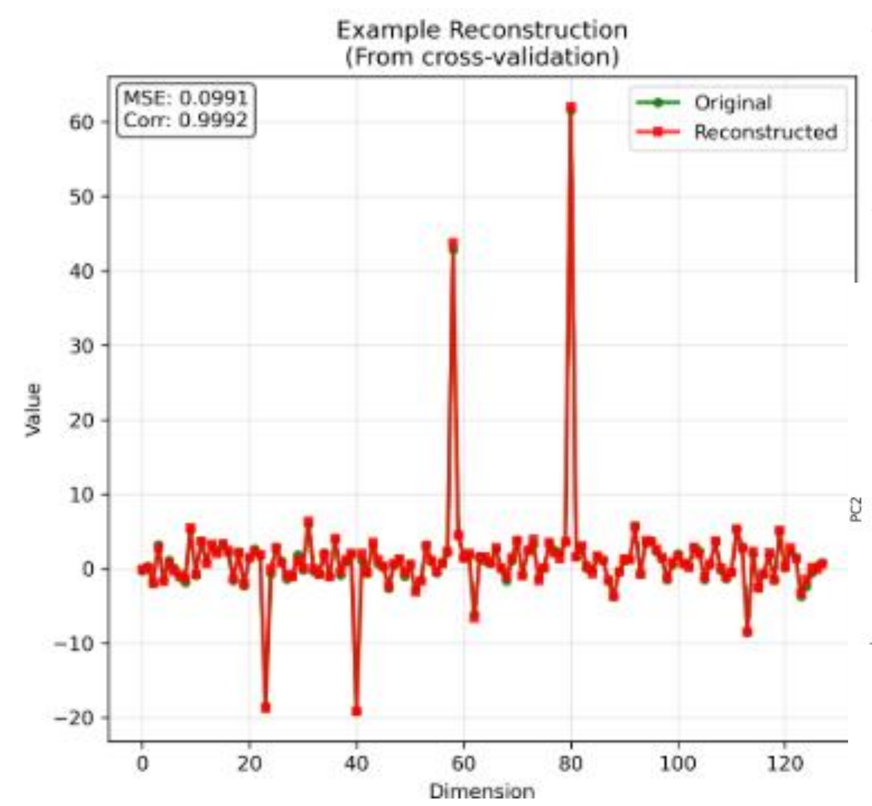
Additional open-source contributions include:
- **SMILES Django Portal (SciGaP)**
  - PR #45: Add sharing functionality
  - PR #48: Fix "select all" logic
  - PR #53 *(under review)*: Add parsing automation script and README
- **Airavata Data Catalog (Apache)**
  - PR #48: Add sharing functionality
  - PR #50: Fix permission for data product detail access

Highlights and Accomplishments – SubProject 1

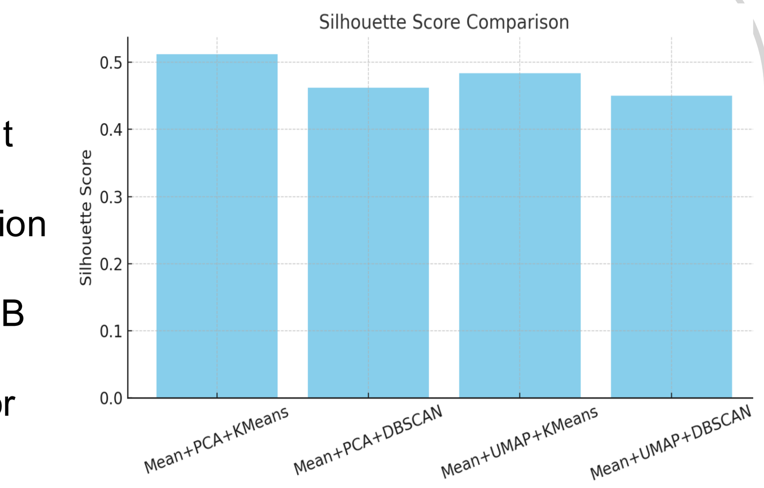


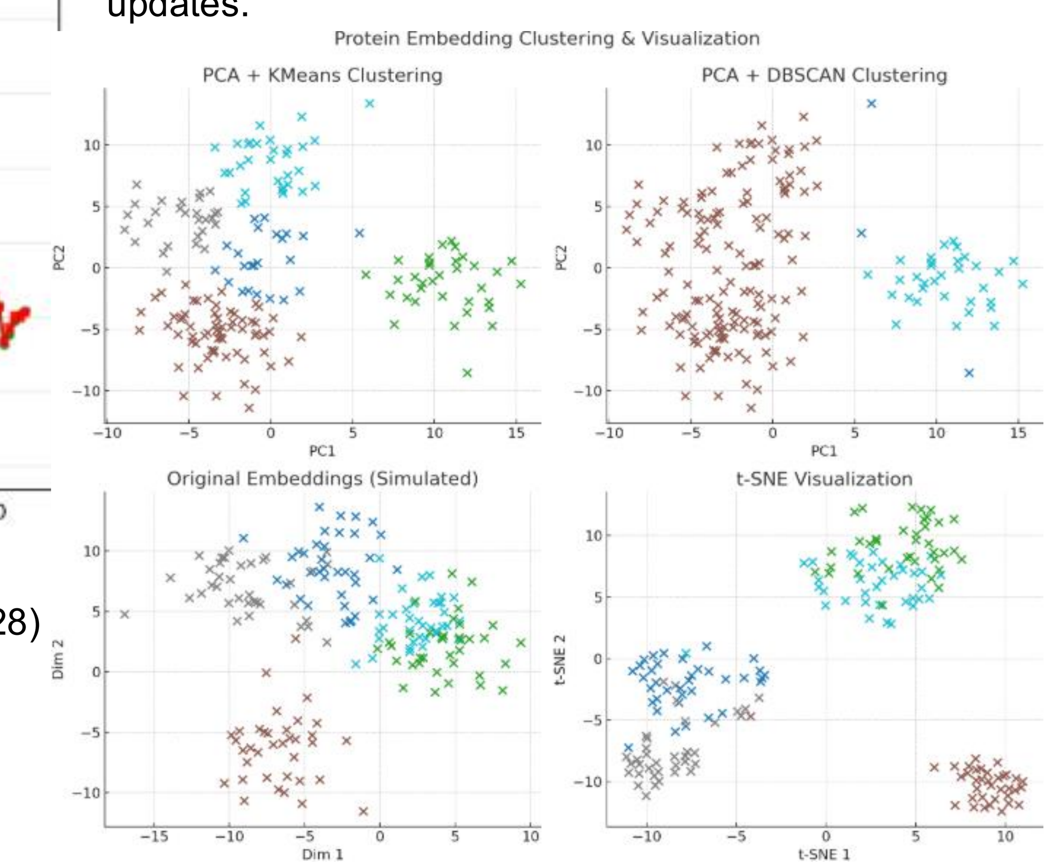
Cross-Validation Performance
- **Mean Test Loss: 0.3223** – Good overall performance
- **90% Success Rate:** 9 out of 10 folds achieve excellent performance (0.1-0.4 range)
- **Consistent Results:** Most folds clustered tightly around the mean
- **Only 1 Problematic Fold:** Fold 7 represents ~10% of dataset with inherent complexity

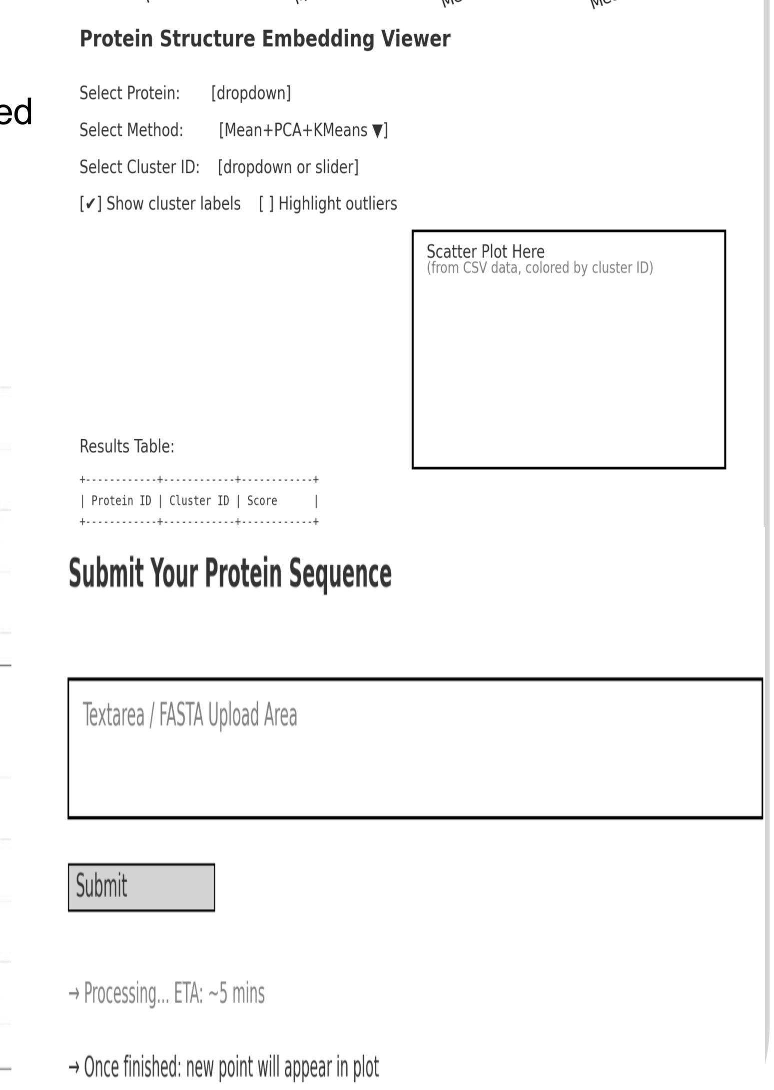
- **MSE: 0.0991** - Extremely low reconstruction error
- **Correlation: 0.9992** - Near-perfect correlation between original and reconstructed embeddings
- **Visual Quality:** The reconstruction plot shows **almost identical overlay** of original (green) and reconstructed (red) protein embeddings across all 128 dimensions



- Ran and verified OpenFold inference locally (PDB output confirmed).
- Built modular scripts for batch inference, pooling, reduction (PCA/UMAP), and clustering (K-means/DBSCAN).
- Evaluated clustering quality using Silhouette, CH, and DB scores.
- Compared 4 method combinations on 180 sequences for best performance.
- Created a lightweight Streamlit viewer for offline result exploration.
- Scaled inference via job arrays on HPC; results integrated seamlessly.
- Outlined future support for real-time input and live plot updates.



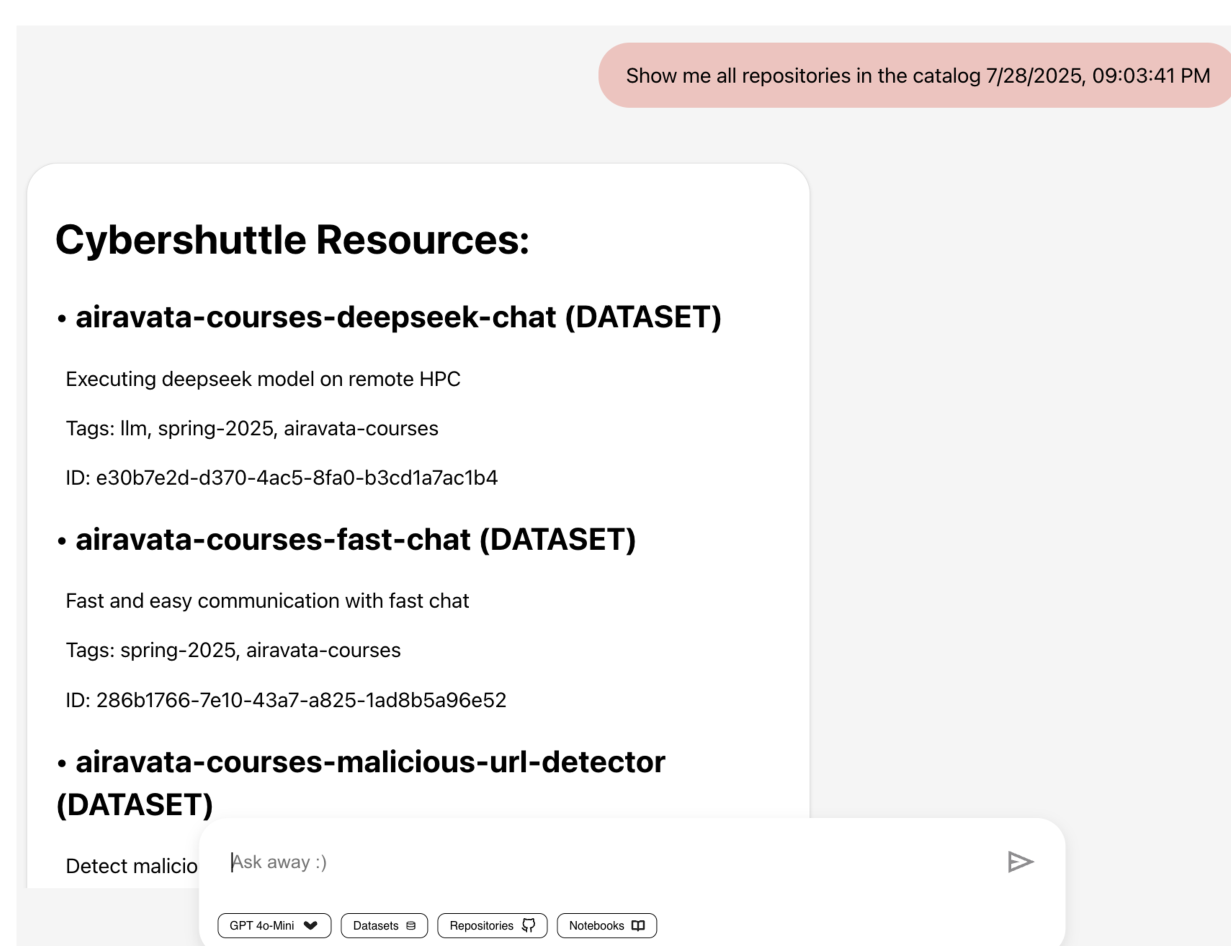




Biological Significance

1. **Protein Representation Learning:** Successfully compressed complex 3D structural information (L×L×128) into meaningful 40-dimensional latent representations
2. **Adaptive Processing:** The AdaptiveAutoencoder learned to distinguish protein complexity automatically, routing simple vs. complex proteins through appropriate pathways
3. **Outlier Detection:** The 10% failure rate likely represents genuinely problematic proteins (membrane proteins, disordered regions, or corrupted data)

Highlights and Accomplishments – SubProject 2



- Created a unified architecture connecting LangChain, OpenAI/GPT-based agents, and Cybershuttle's research catalog.
- Implemented and exposed over a dozen functional MCP-compatible endpoints, including: `GET /resources`, `GET /projects`, `POST /repository`, `GET /tools`, and more.
- Developed a complete `/tools` discovery mechanism for LLM agents to dynamically identify available operations.
- Integrated OAuth2 Device Flow authentication, allowing token-based access to secured Cybershuttle endpoints.
- Designed reusable LangChain tools such as `list_resources`, `get_all_tags`, and `create_repository` using OpenAI function-calling schema.

- Built interactive demo (`cybershuttle_openai_demo.py`) that supports natural language interaction with Cybershuttle datasets and repositories.
- Connected FastAPI-based MCP server to Airavata's backend with live catalog browsing, project creation, and resource filtering.
- Enabled hosting and testing within the Cybershuttle Jupyter-based environment using Airavata's `airavata-python-sdk[notebook]` and GPU runtimes.

Future Work

Some research directions opened up from the subproject1 includes research applicaitons on protein family análisis on latent space clustering to identify protein families, use embeddings to identify similar proteins for target identification, analyze protein evolution through latent space trajectories, compress and analyze large structural databases, etc.
Beyond that, we plan to support real-time single-sequence inference via the Streamlit frontend, integrating on-demand OpenFold runs through HPC backends.

For MCPs, we'll work on add full support for POST endpoints and expand LLM compatibility to open models like Qwen 4B/7B, TinyLLaMA, DeepSeek.

Georgia Tech